12–STABILITY AT DOCKING

12.1 Introduction

To ensure continuous performance of a vessel at the highest level, ships are dry-docked regularly on annual or bi-annually basis to restore the underwater parts to as near new condition as possible.

To avoid the imposition of undue stresses on the hull, dry-docking must be carried out, in as far as possible, the same condition as she was launched when built. To achieve this, she must:

- Be in ballast condition at docking
- Enter docks with the designed stern trim
- Be upright, and
- Be in stable equilibrium (Metacentric height must be positive)

12.2 The Docking Process (Graving Dock)

- On entering dock, a vessel is first aligned accurately over the blocks which have been pre-arranged to suit the ship's bottom plan. Figure 12.1(a)
- The dock gates are then closed and de-watering takes place. This causes the vessel to follow downwards until at the instant the stern frame hits the blocks, a reaction (P) comes into action. Figure 12.1(b)
- As the water level falls and buoyancy is lost, P continues to hold the weight of the ship until when finally, the whole bottom lands, the load is evenly distributed over the total number of blocks. Figure 12.1(c)

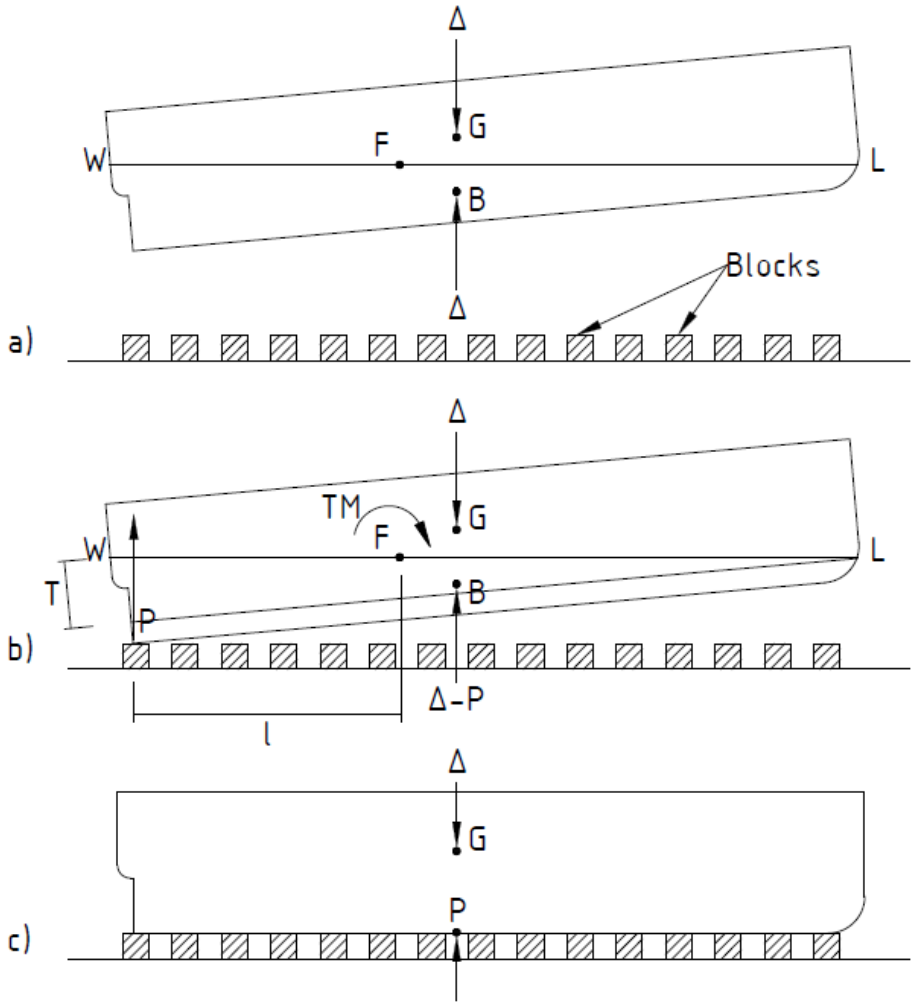


Figure 12.1 – The Docking Process

12.3 Trim at Docking

From the instant a vessel hits the blocks till when trim is reduced to zero, the trimming moment:

$$T = P \times l$$

Now:

$$Change\ in\ trim\ (t) = \frac{TM}{MCT}$$

Therefore:

$$t = \frac{Pl}{MCT}$$

And the load on stern at docking:

$$P = \frac{t.MCT}{l}$$

12.4 Stability at Docking

When viewed transversely, then at the instant a vessel hits the blocks and is subjected to a heel of **θ** degrees, the distribution of forces will be as shown in Figure 12.2.

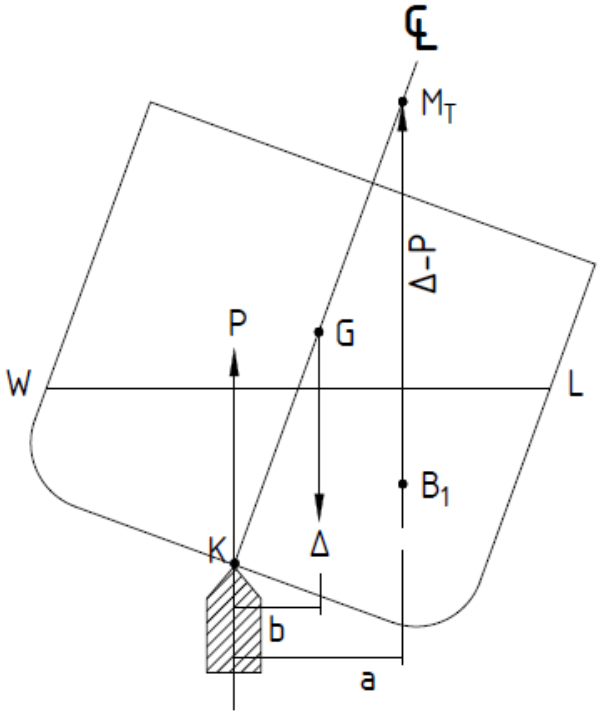


Figure 12.2 – Transverse Stability at Docking

By taking moments about the keel:

$$HM = RM = Anticlockwise\ moment - Clockwise\ moment$$

$$RM = (\Delta - P)a - \Delta b$$

$$RM = (\Delta - P)KM_T sin\theta - \Delta . KG sin\theta$$

Or:

$$RM = \Delta\left(GM_T - \frac{P}{\Delta}KM_T\right)sin\theta$$

The quantity in bracket is therefore the effective metacentric height and is referred to as the ***Virtual Metacentric Height*** at docking. If its value becomes negative anytime during the process the vessel will capsize.

The period between when the stern frame makes contact with the blocks to when the bottom lands overall is referred to as the ***Critical Period*** at docking.

OTHER DOCKING ARRANGEMENTS

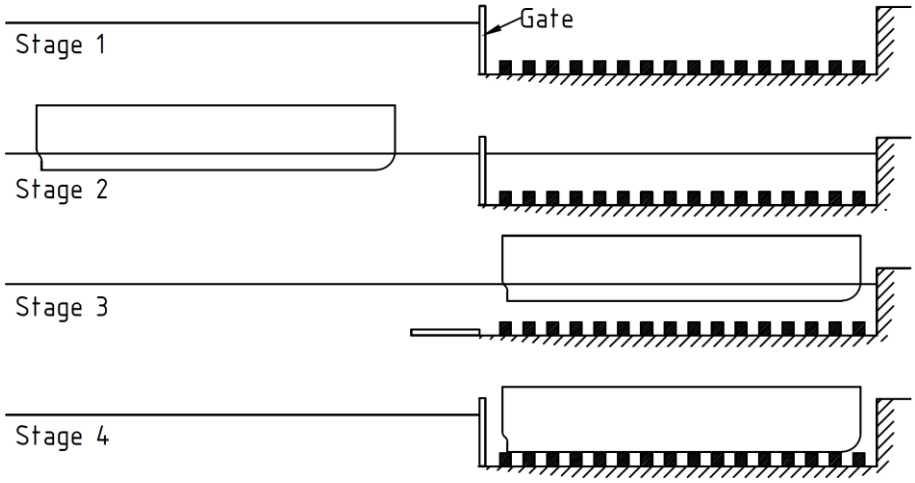


Figure 12.3 – Graving Dock

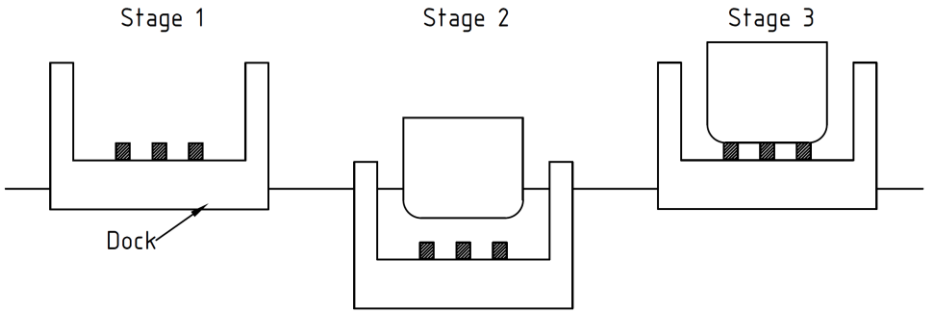


Figure 12.4 – Floating Dock

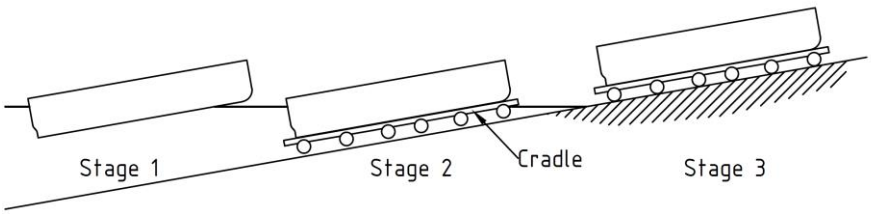


Figure 12.5 – Slipway

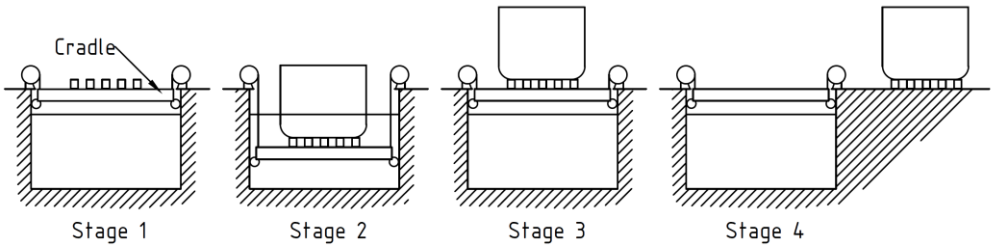


Figure 12.6 – Ship Lift